Relative Standard Error in OLS for A/B Tests

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Motivation

Setup: from absolute to relative effect

We run an A/B test and fit an OLS regression:

$$Y_i = \alpha + \tau \cdot T_i + \mathbf{X}_i^\top \boldsymbol{\beta} + \epsilon_i.$$

- τ = absolute average treatment effect (ATE).
- Stakeholders usually want a relative lift:

$$L \ = \ \frac{\tau}{\bar{Y}_{\text{control}}}.$$

- We need a point estimate *and* a standard error for L.
- Without covariates ($\mathbf{X}_i$ omitted), this OLS is equivalent to the classical two-sample t-test for difference in means.

"divide the SE by the control mean" is a common shortcut

Tempting shortcut:

$$\hat{L} = \frac{\hat{\tau}}{\bar{y}_{\text{control}}}, \qquad \widehat{\mathrm{SE}}(L) \stackrel{?}{=} \frac{\mathrm{SE}(\hat{\tau})}{\bar{y}_{\text{control}}}.$$

Why this is wrong in general:

- The shortcut is the identity

  $$\mathrm{SE}\left(\frac{A}{B}\right) \stackrel{?}{=} \frac{\mathrm{SE}(A)}{B},$$

  which only holds when B is a known constant.
- But $\bar{y}_{\text{control}}$ is an estimate: it has its own standard error.
- Confidence intervals built this way are miscalibrated (too narrow, overly confident).

When does it accidentally work?

The naive formula is approximately correct only when:

- $\mathrm{SE}(\bar{Y}_{\text{control}})/\bar{Y}_{\text{control}} \ll \mathrm{SE}(\hat{\tau})/|\hat{\tau}|$ (control mean is very precisely estimated relative to the effect), and
- $\mathrm{Cov}(\hat{\tau}, \bar{Y}_{\text{control}}) \approx 0$.

In small samples, both assumptions fail. We need the delta method.

Delta method

Delta method: the setup

Suppose we have an estimator $\hat{\boldsymbol{\theta}} \in \mathbb{R}^k$ that is close to its target $\boldsymbol{\theta}$, with a known (or estimated) covariance matrix:

$$\mathrm{Var}(\hat{\boldsymbol{\theta}}) = \Sigma.$$

We care about a smooth scalar transformation $g(\boldsymbol{\theta})$ — e.g. a ratio, a log, a relative lift.

Question. What is $\mathrm{Var}(g(\hat{\boldsymbol{\theta}}))$?

Problem. In general, g is nonlinear, so $\mathrm{Var}(g(\hat{\boldsymbol{\theta}}))$ has no closed form. But if $\hat{\boldsymbol{\theta}}$ is concentrated around $\boldsymbol{\theta}$, we can linearize g and get an approximation.

Step 1: linearize g via Taylor

First-order Taylor expansion of g around the true value $\boldsymbol{\theta}$:

$$g(\hat{\boldsymbol{\theta}}) \;\approx\; g(\boldsymbol{\theta}) \;+\; \nabla g(\boldsymbol{\theta})^\top(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}).$$

Two important things to notice:

- $g(\boldsymbol{\theta})$ is a constant (it's the truth, not random).
- $\nabla g(\boldsymbol{\theta})$ is also a constant vector (gradient evaluated at the truth).
- The only random object on the right-hand side is $\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}$.

So we have reduced $g(\hat{\boldsymbol{\theta}})$ to (a constant) + (a linear function of $\hat{\boldsymbol{\theta}}$).

Step 2: variance of a linear function

For any constant vector $\mathbf{a}$ and random vector $\mathbf{Z}$ with $\mathrm{Var}(\mathbf{Z}) = \Sigma$:

$$\mathrm{Var}(\mathbf{a}^\top \mathbf{Z}) = \mathbf{a}^\top \Sigma\, \mathbf{a}.$$

Quick derivation:

$$\begin{aligned}
\mathrm{Var}(\mathbf{a}^\top \mathbf{Z}) &= \mathbb{E}\big[(\mathbf{a}^\top (\mathbf{Z} - \mu))^2\big] \\
&= \mathbb{E}\big[\mathbf{a}^\top (\mathbf{Z} - \mu)(\mathbf{Z} - \mu)^\top \mathbf{a}\big] \\
&= \mathbf{a}^\top\, \mathbb{E}[(\mathbf{Z} - \mu)(\mathbf{Z} - \mu)^\top]\, \mathbf{a} \;=\; \mathbf{a}^\top \Sigma\, \mathbf{a}.
\end{aligned}$$

We will apply this with $\mathbf{a} = \nabla g(\boldsymbol{\theta})$ and $\mathbf{Z} = \hat{\boldsymbol{\theta}}$.

Step 3: put the two pieces together

From Step 1:

$$g(\hat{\boldsymbol{\theta}}) - g(\boldsymbol{\theta}) \;\approx\; \nabla g(\boldsymbol{\theta})^\top (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}).$$

Taking variance of both sides (the left-hand constant $g(\boldsymbol{\theta})$ drops out):

$$\mathrm{Var}\big(g(\hat{\boldsymbol{\theta}})\big) \;\approx\; \mathrm{Var}\big(\nabla g(\boldsymbol{\theta})^\top \hat{\boldsymbol{\theta}}\big).$$

Applying Step 2 with $\mathbf{a} = \nabla g(\boldsymbol{\theta})$:

$$\boxed{\mathrm{Var}\Big(g(\hat{\boldsymbol{\theta}})\Big) \;\approx\; \nabla g(\boldsymbol{\theta})^\top \,\Sigma\, \nabla g(\boldsymbol{\theta}).}$$

In practice we plug in the estimates $\hat{\boldsymbol{\theta}}$ and $\hat{\Sigma}$ since $\boldsymbol{\theta}$ and Σ are unknown.

Delta method for a ratio

Let $g(X, Y) = X/Y$. Then

$$\nabla g = \begin{pmatrix} 1/Y \\ -X/Y^2 \end{pmatrix}.$$

Plugging into $\nabla g^\top \Sigma \nabla g$:

$$\mathrm{Var}\left(\frac{X}{Y}\right) \approx \frac{\mathrm{Var}(X)}{Y^2} + \frac{X^2}{Y^4}\mathrm{Var}(Y) - 2\,\frac{X}{Y^3}\mathrm{Cov}(X, Y).$$

Here $X = \hat{\tau}$ (numerator) and Y is the estimator of the control mean (denominator). What "Y" is concretely, and how we get $\mathrm{Var}(Y)$ and $\mathrm{Cov}(X, Y)$, depends on whether we have covariates.

Recap

We wanted a standard error for the relative lift

$$\hat{L} \;=\; \frac{\hat{\tau}}{\bar{Y}_{\text{control}}}.$$

The delta-method ratio formula tells us

$$\operatorname{Var}(\hat{L}) \;\approx\; \frac{\operatorname{Var}(\hat{\tau})}{\bar{Y}_{\text{control}}^{\,2}} + \frac{\hat{\tau}^{\,2}}{\bar{Y}_{\text{control}}^{\,4}}\operatorname{Var}(\bar{Y}_{\text{control}}) - 2\,\frac{\hat{\tau}}{\bar{Y}_{\text{control}}^{\,3}}\operatorname{Cov}(\hat{\tau}, \bar{Y}_{\text{control}}).$$

So computing $\operatorname{SE}(\hat{L}) = \sqrt{\operatorname{Var}(\hat{L})}$ has been reduced to finding three ingredients:

1. $\operatorname{Var}(\hat{\tau})$ — variance of the absolute effect,
2. $\operatorname{Var}(\bar{Y}_{\text{control}})$ — variance of the control-mean estimator,
3. $\operatorname{Cov}(\hat{\tau}, \bar{Y}_{\text{control}})$ — their covariance.

The rest of the talk is just: where do these three numbers come from?

Warning

While simple to write, how can we easily calculate $\mathrm{Var}(\bar{Y}_{\text{control}})$ and $\mathrm{Cov}(\hat{\tau}, \bar{Y}_{\text{control}})$? We don't have a closed-form formula for any of them.

Reminder: covariance matrix in OLS

Reminder: OLS hands us the full covariance matrix

We are fitting OLS of the form $Y_i = \alpha + \tau T_i + \mathbf{X}_i^\top \boldsymbol{\beta} + \epsilon_i$.

Model coefficients are $\hat{\boldsymbol{\theta}} = (\hat{\alpha},\, \hat{\tau},\, \hat{\boldsymbol{\beta}}^\top)^\top$. Fitting OLS on the design matrix X with residual variance $\hat{\sigma}^2$ gives:

$$\widehat{\mathrm{Var}}(\hat{\boldsymbol{\theta}}) \;=\; \hat{\sigma}^2 (X^\top X)^{-1} \;=\; \Sigma_{\hat{\theta}}.$$

This is a single $(2+p) \times (2+p)$ matrix containing:

- diagonal: $\mathrm{Var}(\hat{\alpha})$, $\mathrm{Var}(\hat{\tau})$, $\mathrm{Var}(\hat{\beta}_j)$;
- off-diagonal: *every* pairwise covariance, including $\mathrm{Cov}(\hat{\tau}, \hat{\alpha})$ and $\mathrm{Cov}(\hat{\tau}, \hat{\beta}_j)$.

In `statsmodels`: `results.cov_params()` returns exactly this matrix.

As we'll see, both the variance of the denominator and its covariance with $\hat{\tau}$ are going to come out of $\Sigma_{\hat{\theta}}$.

Two cases: no covariates vs. covariates

Case 1: no covariates

Model: $Y_i = \alpha + \tau T_i + \epsilon_i$.

- The control-group mean estimator is simply $\hat{\alpha}$: under $T = 0$, $\mathbb{E}[Y \mid T = 0] = \alpha$, and OLS estimates it with $\hat{\alpha}$.
- So in the ratio $\hat{L} = \hat{\tau}/\hat{\alpha}$, both $X = \hat{\tau}$ and $Y = \hat{\alpha}$ are OLS coefficients.
- OLS already returns the full 2×2 covariance matrix of $(\hat{\alpha}, \hat{\tau})$:

$$\widehat{\mathrm{Var}} \begin{pmatrix} \hat{\alpha} \\ \hat{\tau} \end{pmatrix} = \hat{\sigma}^2 (X^\top X)^{-1}.$$

- So $\mathrm{Var}(\hat{\tau})$, $\mathrm{Var}(\hat{\alpha})$ and $\mathrm{Cov}(\hat{\tau}, \hat{\alpha})$ are read off directly — plug into the ratio formula and we are done.

Case 2: with covariates — the denominator is no longer a coefficient

Model: $Y_i = \alpha + \tau T_i + \mathbf{X}_i^\top \boldsymbol{\beta} + \epsilon_i$.

- Now $\hat{\alpha}$ alone is not an estimate of the control mean. It is the predicted Y when $\mathbf{X} = \mathbf{0}$ — meaningless if covariates are not 0-centered.
- The natural estimator of the control mean is

$$\bar{Y}^{\text{adj}}_{\text{control}} = \hat{\alpha} + \bar{\mathbf{X}}^\top_{\text{control}} \hat{\boldsymbol{\beta}},$$

  i.e. the model's prediction averaged over control covariates.
- This is a linear combination of several OLS coefficients, not a single one.
- Consequences:
  - $\mathrm{Var}(\bar{Y}^{\text{adj}}_{\text{control}})$ is not a single diagonal entry — it is a quadratic form in $\boldsymbol{\Sigma}_{\hat{\theta}}$.
  - $\mathrm{Cov}(\hat{\tau}, \bar{Y}^{\text{adj}}_{\text{control}})$ is not a single off-diagonal entry — it mixes $\mathrm{Cov}(\hat{\tau}, \hat{\alpha})$ with all the $\mathrm{Cov}(\hat{\tau}, \hat{\beta}_j)$.
- We need the machinery of variances of linear combinations.

Linear model: variance of linear combinations

OLS gives us the full covariance matrix

Stack parameters $\boldsymbol{\theta} = (\alpha, \tau, \boldsymbol{\beta}^\top)^\top$. OLS returns:

$$\widehat{\mathrm{Var}}(\hat{\boldsymbol{\theta}}) = \hat{\sigma}^2 (X^\top X)^{-1}.$$

We get every variance *and* every pairwise covariance "for free":

$$\Sigma_{\hat{\theta}} = \begin{pmatrix} \mathrm{Var}(\hat{\alpha}) & \mathrm{Cov}(\hat{\alpha}, \hat{\tau}) & \mathrm{Cov}(\hat{\alpha}, \hat{\boldsymbol{\beta}}^\top) \\ \mathrm{Cov}(\hat{\tau}, \hat{\alpha}) & \mathrm{Var}(\hat{\tau}) & \mathrm{Cov}(\hat{\tau}, \hat{\boldsymbol{\beta}}^\top) \\ \mathrm{Cov}(\hat{\boldsymbol{\beta}}, \hat{\alpha}) & \mathrm{Cov}(\hat{\boldsymbol{\beta}}, \hat{\tau}) & \mathrm{Var}(\hat{\boldsymbol{\beta}}) \end{pmatrix}.$$

This is the key object that makes a clean delta-method calculation possible.

Variance of a linear combination of parameters

For any constant vector $\mathbf{c}$:

$$\mathrm{Var}(\mathbf{c}^\top \hat{\boldsymbol{\theta}}) = \mathbf{c}^\top \Sigma_{\hat{\theta}} \mathbf{c}.$$

Expanding component-wise for $\mathbf{c}^\top \hat{\boldsymbol{\theta}} = \sum_j c_j \hat{\theta}_j$:

$$\mathrm{Var}\left(\sum_j c_j \hat{\theta}_j\right) = \sum_j c_j^2 \mathrm{Var}(\hat{\theta}_j) + 2 \sum_{j<k} c_j c_k \,\mathrm{Cov}(\hat{\theta}_j, \hat{\theta}_k).$$

And for the covariance with another linear combination $\mathbf{d}^\top \hat{\boldsymbol{\theta}}$:

$$\mathrm{Cov}(\mathbf{c}^\top \hat{\boldsymbol{\theta}}, \mathbf{d}^\top \hat{\boldsymbol{\theta}}) = \mathbf{c}^\top \Sigma_{\hat{\theta}} \mathbf{d}.$$

Relative lift in OLS

The adjusted control mean

With covariates, the predicted outcome for a control unit with covariates $\mathbf{x}$ is

$$\hat{\mathbb{E}}[Y \mid T = 0, \mathbf{X} = \mathbf{x}] = \hat{\alpha} + \mathbf{x}^\top \hat{\boldsymbol{\beta}}.$$

Averaging over the empirical covariate distribution of the control group:

$$\boxed{\bar{Y}^{\text{adj}}_{\text{control}} = \hat{\alpha} + \bar{\mathbf{X}}^\top_{\text{control}} \hat{\boldsymbol{\beta}}.}$$

This is a linear combination $\mathbf{c}^\top \hat{\boldsymbol{\theta}}$ with

$$\mathbf{c} = (1,\ 0,\ \bar{\mathbf{X}}^\top_{\text{control}})^\top \quad \text{(weight on } \hat{\tau} \text{ is 0).}$$

Each piece of the delta-method formula

Same three ingredients we identified earlier, now written for the adjusted control mean $\bar{Y}^{\text{adj}}_{\text{control}}$ in the denominator:

$$\operatorname{Var}(\hat{L}) \approx \underbrace{\frac{\operatorname{Var}(\hat{\tau})}{(\bar{Y}^{\text{adj}}_{\text{control}})^2}}_{\text{(A) Var}(\hat{\tau})} + \underbrace{\frac{\hat{\tau}^2}{(\bar{Y}^{\text{adj}}_{\text{control}})^4}\operatorname{Var}(\bar{Y}^{\text{adj}}_{\text{control}})}_{\text{(B) Var}(\bar{Y}^{\text{adj}})} - \underbrace{2\,\frac{\hat{\tau}}{(\bar{Y}^{\text{adj}}_{\text{control}})^3}\operatorname{Cov}(\hat{\tau}, \bar{Y}^{\text{adj}}_{\text{control}})}_{\text{(C) Cov}(\hat{\tau}, \bar{Y}^{\text{adj}})}.$$

We now compute each of (A), (B), (C) as a function of entries of $\Sigma_{\hat{\theta}}$.

(A) Variance of $\hat{\tau}$

Read it off the OLS covariance matrix.

$$\mathrm{Var}(\hat{\tau}) = \left[\Sigma_{\hat{\theta}}\right]_{\tau,\tau}.$$

This is what `statsmodels` returns as `bse[treatment]**2`.

(B) Variance of $\bar{Y}^{\text{adj}}_{\text{control}}$

Quadratic form with $\mathbf{c} = (1, \bar{\mathbf{X}}_{\text{control}}^\top)^\top$ on the $(\alpha, \boldsymbol{\beta})$ block:

$$\operatorname{Var}(\bar{Y}^{\text{adj}}_{\text{control}}) = \begin{pmatrix} 1 & \bar{\mathbf{X}}_{\text{control}}^\top \end{pmatrix} \operatorname{Var}\begin{pmatrix} \hat{\alpha} \\ \hat{\boldsymbol{\beta}} \end{pmatrix} \begin{pmatrix} 1 \\ \bar{\mathbf{X}}_{\text{control}} \end{pmatrix}.$$

Expanded:

$$\operatorname{Var}(\hat{\alpha}) + \bar{\mathbf{X}}_{\text{control}}^\top \operatorname{Var}(\hat{\boldsymbol{\beta}})\, \bar{\mathbf{X}}_{\text{control}} + 2\operatorname{Cov}(\hat{\alpha}, \hat{\boldsymbol{\beta}})^\top \bar{\mathbf{X}}_{\text{control}}.$$

Every term lives in $\Sigma_{\hat{\theta}}$.

(C) Covariance of $\hat{\tau}$ with the adjusted control mean

$\hat{\tau}$ is linear in $\hat{\boldsymbol{\theta}}$ (with weight 1 on τ, 0 elsewhere); the control mean has weights $(1, \bar{\mathbf{X}}_{\text{control}})$ on $(\alpha, \boldsymbol{\beta})$. So:

$$\operatorname{Cov}(\hat{\tau},\ \bar{Y}^{\text{adj}}_{\text{control}}) = \operatorname{Cov}(\hat{\tau}, \hat{\alpha}) + \sum_j \bar{X}_{\text{control},j} \operatorname{Cov}(\hat{\tau}, \hat{\beta}_j).$$

MDE for relative lift

What about relative MDE

For an absolute effect, MDE at significance α and power $1-\beta$ is:

$$\mathrm{MDE}_{\mathsf{abs}} = (z_{1-\alpha/2} + z_{1-\beta}) \cdot \mathrm{SE}(\hat{\tau}).$$

For relative lift one is tempted to write:

$$\mathrm{MDE}_{\mathsf{rel}} \stackrel{?}{=} (z_{1-\alpha/2} + z_{1-\beta}) \cdot \mathrm{SE}(\hat{L}).$$

But $\mathrm{SE}(\hat{L})$ itself depends on the true lift — this formula is circular. The correct way to solve for $\mathrm{MDE}_{\mathsf{rel}}$ is to plug in the formula for $\mathrm{SE}(\hat{L})$ and solve the resulting quadratic equation. Luiz is working on this for the delta method, but looking for contributor for OLS.

This means that relative MDE sometimes does not exist.

In cluster_experiments

All of this is wired up via `relative_effect`

Everything we derived (delta-method SE, adjusted control mean) is implemented behind a single flag:

```
relative_effect: True
```

- Available for `OLSAnalysis` and `ClusteredOLSAnalysis` (it needs the full $\Sigma_{\hat{\theta}}$).
- Switches the analyser to report $\hat{L} = \hat{\tau}/\bar{Y}^{\text{adj}}_{\text{control}}$ with delta-method SE instead of the raw $\hat{\tau}$.
- Carries through to `NormalPowerAnalysis`: `.mde(...)` returns the relative MDE, not the absolute one (not fully correct, does not solve the quadratic yet).

Config example 1: vanilla OLS, no covariates

```
config = {
    "analysis": "ols",
    "splitter": "non_clustered",
    "target_col": "orders",
    "relative_effect": True,
}

pw = NormalPowerAnalysis.from_dict(config)
pw.mde(user_df, n_simulations=1)   # -> relative MDE
```

Here $Y = \hat{\alpha}$: Case 1 from earlier. Internally we use the 2×2 covariance block of $(\hat{\alpha}, \hat{\tau})$.

Config example 2: OLS with covariates

```
config = {
    "analysis": "ols",
    "splitter": "non_clustered",
    "target_col": "orders",
    "covariates": ["X1", "X2"],
    "relative_effect": True,
}

pw = NormalPowerAnalysis.from_dict(config)
pw.mde(user_df, n_simulations=1)
```

Here $Y = \hat{\alpha} + \bar{\mathbf{X}}_{\text{control}}^{\top}\hat{\boldsymbol{\beta}}$: Case 2. The full covariance matrix of $(\hat{\alpha}, \hat{\tau}, \hat{\boldsymbol{\beta}})$ is used to build $\mathrm{Var}(Y)$ and $\mathrm{Cov}(\hat{\tau}, Y)$.

Config example 3: `AnalysisPlan` on real experiment data

Same flag, but on the inference side (analysing data from a finished test):

```
config = {
    "metrics": [
        {"alias": "Orders", "name": "orders"},
    ],
    "variants": [
        {"name": "A", "is_control": True},
        {"name": "B", "is_control": False},
    ],
    "analysis_type": "ols",
    "variant_col": "treatment",
    "analysis_config": {"relative_effect": True, "covariates": ["X1"]},
}

results = AnalysisPlan.from_metrics_dict(config).analyze(user_df)
```

`results.ate` is the relative lift $\hat{L}$; `results.std_error` is the delta-method SE; CIs and p-values follow.

Live demo

End-to-end runnable example (data simulation, absolute vs. relative SE, absolute vs. relative MDE):

cluster_experiments docs — relative effects notebook

https://david26694.github.io/cluster-experiments/relative.html#experiment-analysis-with-absolute-and-relative-effects

Takeaways

- Relative lift is a ratio of estimators; treat it as such.
- Delta method + full OLS covariance matrix gives the correct SE in one clean formula.
- For MDE estimation, solve the quadratic — linear scaling is only an approximation valid when the control mean is essentially noise-free.
- In `cluster_experiments`, flip `relative_effect: True` on `OLSAnalysis`/`ClusteredOLSAnalysis` — it works for both `NormalPowerAnalysis` (planning, relative MDE) and `AnalysisPlan` (inference, relative ATE + SE).